A Benchmark Generator of Realistic Logistic Transport Networks with Nonlinear Edge Costs and Transshipment Capacities

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Abstract

Optimisation over logistic transport networks—routing many commodities at minimum joint cost of vehicle movements and transshipment at intermediate storages—lacks a public benchmark when edge costs are *nonlinear* (freight travels in fixed-capacity vehicles) and storages have *hard reload-capacity* limits. We close this gap with a generator of synthetic logistic networks, calibrated on real road graphs of nine European countries, five U.S. states and the European part of Russia extracted from OpenStreetMap. The generator reproduces the structural signatures of upper-level distribution networks—near-planar topology, low vertex degree, short edges and large diameters—by combining four well-grounded ingredients: Zipf-distributed city sizes, a spatial-network construction model that trades edge-building cost against routing convenience, storage capacities driven by population and betweenness centrality, and a doubly constrained gravity model for demand. We release 34 fully synthetic instances (10–150 storages) and 15 real-geometry instances, in an open, self-describing file format. An empirical characterisation, computed directly from the released files, shows that the synthetic instances match the real ones across degree, diameter, edge-length, rank–size and demand statistics, confirming that the corpus is a faithful and reusable testbed for exact and heuristic solvers.

Keywords: benchmark generation; logistic networks; spatial networks; multi-commodity flow; gravity model; Zipf's law; synthetic data.

1 Introduction

Planning commodity flows between distribution centres is a core problem for logistic operators, and its realistic variants are computationally hard. Two features make real instances especially difficult. First, the cost of using a road is *nonlinear* in the volume carried: goods move in vehicles of fixed capacity, so the billed quantity is the integer number of vehicles obtained by rounding the carried volume up to a vehicle capacity. Second, every storage has a finite *transshipment capacity*, an upper bound on the volume that can be unloaded, stored and reloaded there. The resulting model is a capacitated multi-commodity flow problem with a step-wise vehicle cost—NP-hard, and sensitive to the topology and the demand pattern of the underlying network.

Progress on solvers for such problems is hampered by the absence of suitable public benchmarks. Classical multi-commodity flow libraries do not carry node-level transshipment capacities, per-unit reload costs or vehicle-induced nonlinear edge costs, and they rarely reflect the geographic structure of country-scale distribution networks. Moreover, the true locations of commercial distribution

centres are proprietary. Researchers are therefore left to test on either toy graphs or ad-hoc random graphs whose structure differs sharply from real transport networks, which undermines the external validity of any comparison.

Contributions. This paper introduces a generator and an accompanying instance corpus designed to remove that obstacle. Specifically, we

1. define precisely the data an instance of the problem must contain (Section 2);
2. analyse real road graphs extracted from OpenStreetMap and distil the structural properties a realistic generator must reproduce (Section 3);
3. present a five-stage generation pipeline combining Zipf city sizes, a spatial-network budget model, centrality-driven storage capacities, and a gravity demand model (Section 4);
4. describe the released benchmark and its open file format (Section 5); and
5. characterise the corpus empirically, directly from the released files, showing that synthetic and real-geometry instances are statistically consistent (Section 6).

The optimisation algorithm we developed for this problem is reported separately; here the focus is the data and its fidelity.

2 The optimisation problem an instance must support

To fix the data requirements, we briefly state the target problem. The network is a directed graph $G = (V, E)$ of storages V and roads E. A set of transportation requests (commodities) K is given; each $k \in K$ has an origin s_k, a destination t_k and a demand $d_k \in \mathbb{R}_+$. Each storage v has a per-unit reload cost $f_v \in \mathbb{R}_+$ and a maximum transshipment volume $W_v \in \mathbb{R}_+$; each edge e has a per-vehicle traversal cost $c_e \in \mathbb{R}_+$; and vehicles have a fixed capacity C. Writing x_e^k for the flow of commodity k on edge e and $y_e \in \mathbb{Z}_+$ for the number of vehicles on e, the problem minimises movement plus reload cost,

$$\min \sum_{e \in E} c_e \, y_e \; + \sum_{k \in K} \sum_{v \neq s_k,\, v \neq t_k} \sum_{e \in \delta^-(v)} f_v \, x_e^k, \tag{1}$$

subject to flow conservation, the vehicle-capacity coupling $\sum_k x_e^k \leq C \, y_e$ for all e, and the storage-capacity bound $\sum_k \sum_{e \in \delta^-(v)} x_e^k \leq W_v$ for all transit v.

Consequently, a complete instance must specify: a weighted graph with vertex coordinates; per-storage reload cost f_v and capacity W_v; per-edge cost c_e (here proportional to road length); a vehicle capacity C; and an origin–destination demand set $\{(s_k, t_k, d_k)\}$. The generator below produces exactly these objects, together with the auxiliary data (populations, shortest distances) used to derive them.

3 Structural properties of real networks

Since the locations of real distribution centres are not public, we calibrate on city locations, populations and highways from OpenStreetMap [1], for nine European countries of varying size, five large U.S. states, and the European part of Russia. From the raw highway graph and city list we aggregate nearby road endpoints into nodes, select cities greedily by descending population subject to a minimum spacing, attach each retained city to a nearby high-degree road node (junctions are

Table 1: Hop diameter and size of representative calibration graphs.

Network	diam	$\|V\|$	Network	diam	$\|V\|$
Belgium	5	16	Italy	11	46
Denmark	5	11	Sweden	11	45
Germany	8	49	United Kingdom	9	42
Spain	9	59	California	12	39
Europe Russia	16	90	Texas	10	58

economical places for a centre), and finally simplify the graph so that each retained centre is joined to others by edges equal to the shortest stop-free paths. Edges are then treated as straight segments of Euclidean length, as modelling the true road geometry would over-complicate synthesis without materially changing the network structure.

Three structural signatures emerge robustly and become the design targets of the generator. (i) The edge-length distribution is dominated by short edges, with very few long edges between distant nodes. (ii) Hop diameters are large relative to size (Table 1); together with the edge-length distribution this indicates the *opposite* of a small-world structure—routes between distant nodes pass through many short edges. (iii) The mean vertex degree is low: about 3.4 across all calibration graphs, well below the planar upper bound of 6, so the networks are essentially planar (drawable without edge crossings). These properties are characteristic of spatial transport networks [4] and any plausible generator must reproduce them.

4 The generation pipeline

Given a desired number of storages k, the generator proceeds in five stages.

4.1 City sizes via Zipf's law

A pool of cities (several times larger than k) is created with sizes following Zipf's law [2],

$$size_i = \frac{P}{rank_i^{\alpha}}, \qquad rank_i = 1, 2, \ldots, N, \tag{2}$$

where P is the largest city's size and small multiplicative noise is added for realism. Across the calibration regions the fitted exponent clusters near $\alpha \approx 0.974$, so the generator draws $\alpha \sim \mathcal{N}(0.974, 0.06)$, the largest population from $(9, 11) \cdot 10^4\, k$, and a city count of k times a random integer in $[8, 12]$.

4.2 Spatial layout

The k city clusters are placed on a square map with `make_blobs` (cluster standard deviation about 1/15 of the map side), seeding the largest city in each cluster; this imitates the real pattern of smaller towns surrounding a large city. Nearby cities are then aggregated into storages exactly as in the calibration step, and each storage inherits the summed population of its nearest cities. The minimum spacing between storages is a random value in $[80, 100]$ and the map side scales as that spacing times $k^{0.8}$.

4.3 Topology via a spatial-network budget model

Following Gastner and Newman [3], edges are chosen to balance construction cost against routing convenience. Construction cost is the total edge length $\sum_{(i,j)} d_{ij}$ (Euclidean, in km), while the routing "effective distance" of an edge is

$$L_{ij} = \lambda\, d_{ij} + (1-\lambda)\cdot 1, \qquad 0 \le \lambda \le 1, \tag{3}$$

so that $\lambda = 1$ measures routes in kilometres and $\lambda = 0$ in hop count. Starting from a minimum spanning tree (which both guarantees connectivity and sets the budget reference), edges are added within a budget—a coefficient times the MST cost—so as to minimise mean pairwise effective distance, an optimisation we solve by simulated annealing. Matching the diameter, mean edge length and mean degree of the calibration graphs yields $\lambda \sim \mathcal{N}(0.263, 0.15)$ and budget coefficient $\sim \mathcal{N}(2.56, 0.6)$: a small budget admits only a few long edges among many short ones, exactly the observed real pattern. Each edge is duplicated in reverse with length differing by at most one percent, and per-vehicle cost is $c_e = cost_{km} \cdot dist_e$ with $cost_{km} = 5$.

4.4 Storage capacities and reload costs

A storage's transshipment capacity should reflect both the demand it serves (population) and the through-flow it attracts. The latter correlates strongly with betweenness centrality [4, 5], so capacity is

$$maxoverload_i = l \cdot population_i^{\gamma}\,(1 + \alpha\,\tilde{b}_i), \tag{4}$$

where $\tilde{b}_i$ is normalised betweenness, with $l = 1, \gamma = 0.5, \alpha = 1.2$ and a floor $mincapacity = 100$ (capacities are adapted so at least 10% fall below the floor). Reload costs decrease with capacity, reflecting economies of scale:

$$cost_i = \frac{\kappa}{maxoverload_i^{\delta}}, \qquad \kappa = 500,\ \delta = 1. \tag{5}$$

4.5 Demand via a doubly constrained gravity model

The origin–destination matrix uses a doubly constrained gravity model [6],

$$T_{ij} = A_i\, O_i\, B_j\, D_j\, f(dist_{ij}), \qquad f(dist) = dist^{-\beta},\ \beta = 1, \tag{6}$$

where the marginals O_i, D_j are proportional to served population (plus small noise), $dist_{ij}$ is the shortest network distance, and the balancing factors A_i, B_j enforce $\sum_j T_{ij} = O_i$ and $\sum_i T_{ij} = D_j$, computed by iterative fitting. Total flow is scaled to a few times the total storage capacity so that the capacity constraints bind. Finally 25% of requests are dropped with probability proportional to their volume, so that not every pair exchanges goods, as in real demand matrices.

5 The benchmark instances

The corpus has two parts. The **synthetic** set, produced fully by the pipeline, contains 34 instances: 10 with 10 storages, 7 each with 20 and 30, 5 with 50, 3 with 100 and 3 with 150. The **real-geometry** set contains 15 instances built on the calibration regions (11 to 90 storages): the vertex coordinates and populations come from OpenStreetMap, while edge costs, storage capacities, reload

Table 2: File format of a benchmark instance.

File	Contents
`graph.graphml`	weighted graph; node `x,y`; edge `weight` (road length)
`offices.csv`	`office_id, transfer_price` (f_v), `transfer_max` (W_v)
`pos_pop.csv`	city/storage `x, y, population`
`reqs.csv`	demand requests `src, dst, volume` (s_k, t_k, d_k)
`distance_matrix.csv`	all-pairs shortest road distances
`data.txt`	random seed and all generation parameters

Table 3: Synthetic instances aggregated by size (n instances per size): mean degree, hop diameter, mean edge length, mean storage capacity $\bar{W}$, mean reload cost $\bar{f}$, number of commodities $|K|$ and mean demand $\bar{d}$.

$\lvert V\rvert$	n	deg_{mean}	diam	$dist_{\text{mean}}$	$\bar{W}$	$\bar{f}$	$\lvert K\rvert$	$\bar{d}$
10	10	3.68	3.3	189.5	266.0	2.63	67	153.4
20	7	4.06	5.3	261.6	230.4	2.75	285	56.7
30	7	3.25	7.9	261.9	199.1	3.00	652	36.1
50	5	3.18	9.4	304.8	182.8	3.22	1837	21.9
100	3	3.05	12.0	481.0	178.6	3.27	7425	17.2
150	3	3.15	18.7	477.0	177.4	3.30	16762	12.4

costs and demand are generated by stages 3–5 above. Together this spans 10 to 150 storages and 67 to over 16,000 commodities.

Each instance is a directory of plain, self-describing files (Table 2): an undirected weighted graph in GraphML with vertex coordinates, a per-storage table of reload cost and capacity, a city position/population table, the demand requests, a precomputed shortest-distance matrix, and a parameter file recording the random seed and every generation parameter, so each instance is exactly reproducible.

6 Empirical characterisation

We now characterise the corpus using statistics computed directly from the released files, to verify that the synthetic instances reproduce real network structure. Tables 3 and 4 summarise the per-size synthetic aggregates and the individual real networks; all figures below pool every instance unless stated otherwise.

Topology. Figure 1 shows a synthetic instance beside the Spain network: both display the same near-planar, locally clustered structure with mostly short edges. The degree distribution (Figure 2a) is tightly concentrated below the planar bound of 6, with synthetic and real means of 3.4 and 3.5 respectively. The edge-length distribution (Figure 2b), normalised by each graph's largest pairwise distance, is right-skewed and dominated by short edges in both populations. Hop diameter grows with size faster than the $\sqrt{|V|}$ reference (Figure 3a), confirming the non-small-world character; synthetic and real points overlap throughout.

Populations and instance parameters. The rank–size relation of storage populations (Figure 3b) is approximately log-linear on log–log axes for both a synthetic instance and Spain, con-

Table 4: Real-geometry instances: structural and instance statistics computed from the released files. W_{sum} is total storage capacity, d_{sum} total demand.

Network	$\lvert V\rvert$	deg_{mean}	diam	$dist_{\text{mean}}$	W_{sum}	$\bar{f}$	$\lvert K\rvert$	d_{sum}
Denmark	11	2.55	5	130.4	3294	2.35	82	8323
Belgium	16	3.12	5	76.3	5703	2.11	180	15717
Netherlands	18	3.67	5	97.3	4694	2.73	229	11605
Hungary	19	2.74	7	103.3	4566	3.26	256	20566
Ohio USA	22	3.73	6	106.8	7371	2.58	346	22994
Georgia USA	24	3.75	7	112.4	7720	2.57	414	34832
Illinois USA	28	3.50	6	109.5	10292	2.53	567	24337
California USA	39	2.56	12	138.5	36544	1.48	1111	114359
United Kingdom	42	4.19	9	171.2	24109	1.75	1291	77119
Sweden	45	3.42	11	225.4	21373	1.83	1485	85688
Italy	46	3.61	11	134.7	14288	2.40	1552	61953
Germany	49	5.39	8	206.8	14276	2.54	1764	43705
Texas USA	58	3.00	10	138.5	27767	2.31	2479	110540
Spain	59	4.14	9	151.5	19789	2.38	2566	77666
Europe Russia	90	4.27	16	394.2	42492	2.03	6007	132570

sistent with the Zipf prior used in stage 1. Storage capacities and reload costs follow the intended inverse relationship—larger storages reload more cheaply (Figure 4, left)—and request volumes are right-skewed with many small and few large shipments (Figure 4, right), as produced by the gravity model.

Consistency. Across every statistic examined—degree, diameter scaling, edge-length shape, rank–size slope, and the capacity/demand distributions—the synthetic and real-geometry instances behave alike (cf. Tables 3–4 and Figures 2–4). This is the central validation result: the generator does not merely produce hard instances, it produces instances whose structure is statistically indistinguishable, in the measured respects, from networks built on real geography. Solvers tuned on the synthetic set can therefore be expected to transfer to real-world inputs.

7 Reproducibility and tooling

Every instance records its random seed and full parameter set in `data.txt`, so the corpus regenerates exactly. To inspect instances and solver outputs we provide an interactive web application (Streamlit/Plotly) that loads any instance from the corpus and draws the network on its geographic layout, with node size encoding population and capacity, edge width encoding vehicle load, and per-node congestion highlighted. The tool supports per-commodity path inspection and aggregate statistics, and was used to produce the layout views in this paper. The generator, the corpus and the viewer are distributed together so that results on this benchmark are fully reproducible.

8 Conclusion

We have presented a generator and an open instance corpus for logistic network flow problems with nonlinear, vehicle-induced edge costs and hard storage transshipment capacities—a problem class for which no realistic public benchmark previously existed. By calibrating on real road graphs and

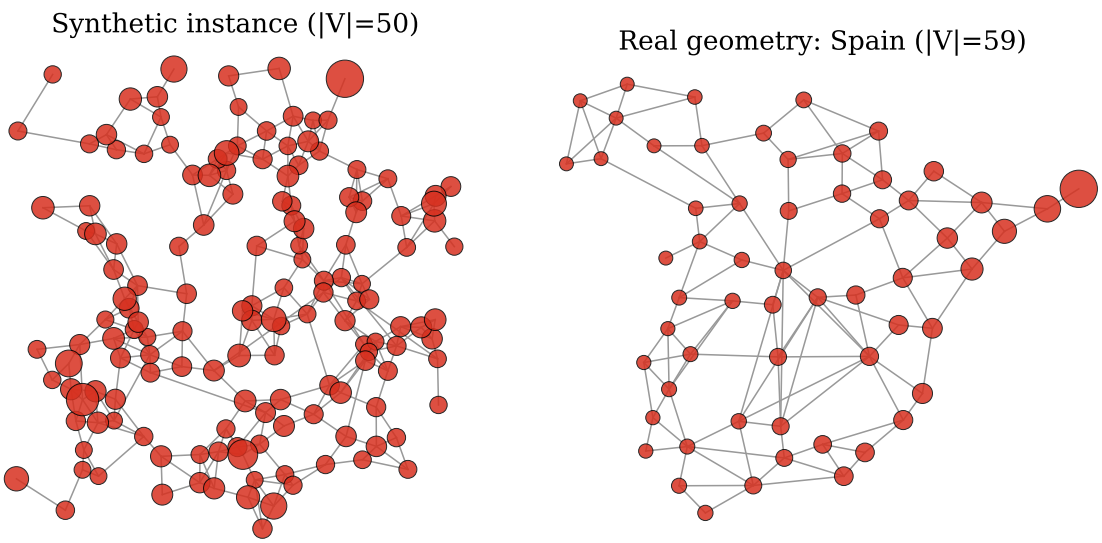


Figure 1: Generated layouts. Left: a fully synthetic instance ($|V| = 50$). Right: a real-geometry instance (Spain, $|V| = 59$). Node area encodes storage capacity. Both share the near-planar, short-edge structure of real transport networks.

combining Zipf city sizes, a spatial-network budget model, centrality-driven capacities and a gravity demand model, the generator reproduces the near-planar topology, low degree, short edges and large diameters of real upper-level distribution networks. An empirical characterisation computed directly from the released files shows that the 34 synthetic and 15 real-geometry instances are statistically consistent across all measured properties, making the corpus a faithful and reusable testbed. Natural extensions include time-dependent demand, multi-modal edges, and stochastic capacities, each of which the parameterised pipeline can accommodate.

9 Acknowledgments.

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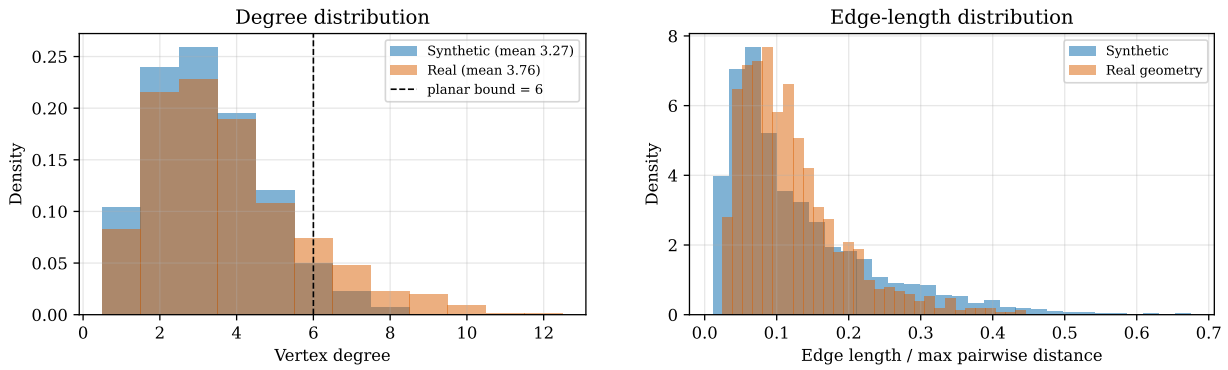
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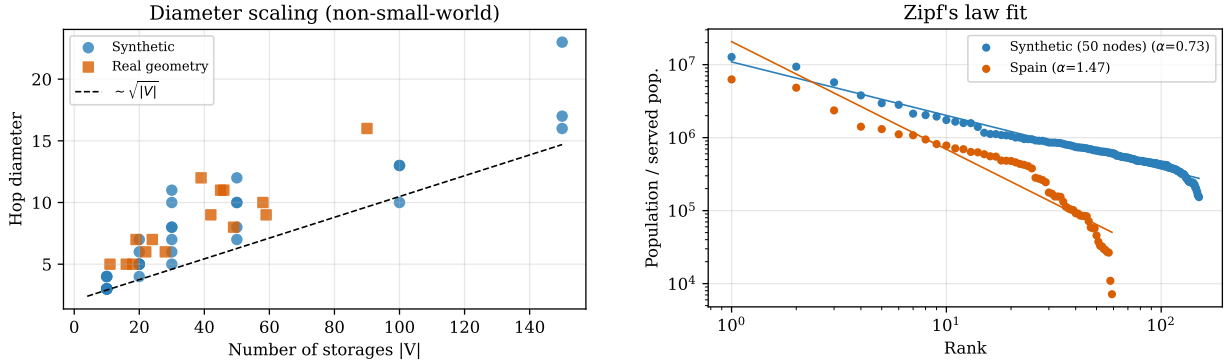
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(a) Degree distribution.

(b) Edge-length distribution.

Figure 2: Pooled topological distributions over all instances. Synthetic and real-geometry instances are closely aligned, and both stay below the planar degree bound and are dominated by short edges.



(a) Diameter vs. size.

(b) Rank–size (Zipf) relation.

Figure 3: Left: hop diameter grows faster than $\sqrt{|V|}$ for both populations, i.e. networks are not small-world. Right: the rank–size relation of storage populations is approximately log-linear, as Zipf's law prescribes.

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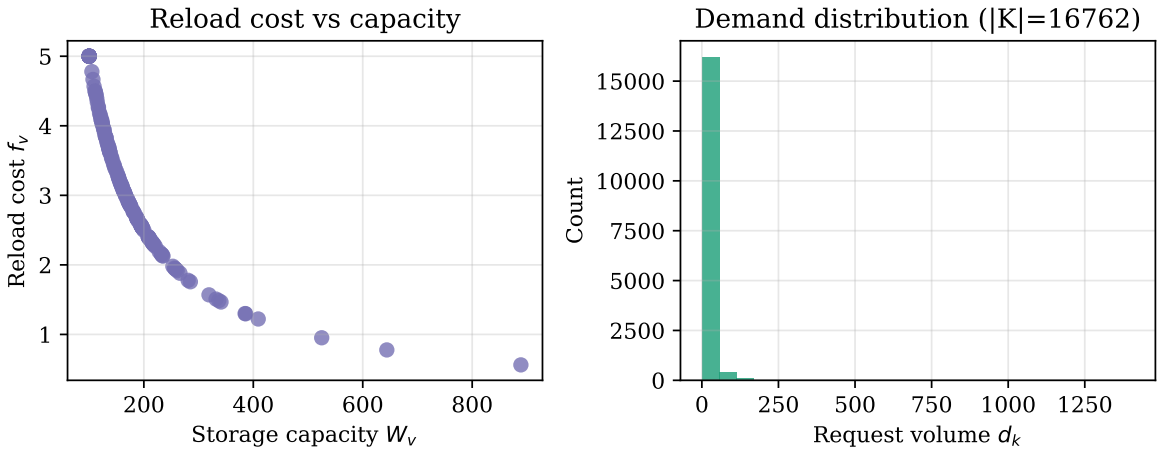


Figure 4: Instance parameters for a synthetic $|V| = 50$ network. Left: reload cost f_v decreases with storage capacity W_v (economies of scale). Right: right-skewed distribution of request volumes d_k from the gravity model.